

UDIT PANDEY

FULL STACK DEVELOPER | AI ENGINEER | ROBOTICS INNOVATOR

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EDUCATION

B.Tech in Computer Science and Engineering

2025 – 2029

Institute of Technology and Management

Senior Secondary & Secondary Education

Cambridge School, Dabra

TECHNICAL SKILLS

Web Development: React, Next.js (15+), TypeScript, JavaScript, Tailwind CSS, Shadcn UI, Full Stack Engineering, UI/UX Design

Artificial Intelligence: Machine Learning, Deep Learning, Neural Networks

Computer Vision: OpenCV, Face Recognition, Gesture Recognition

Robotics & IoT: Embedded Systems, Automation, Sensor Integration, ESP32, Arduino, Wireless Communication

Tools: Git, GitHub, VS Code, Vercel

TECHNICAL PROJECTS

Mark 50 Iron Man Helmet

IoT & Robotics Innovation

Technologies: Embedded Systems, Automation, Sensor Integration, ESP32, Arduino, Wireless Communication

- Designed and fabricated a realistic mechanical hardware prototype inspired by the Marvel Iron Man chassis.
- Integrated automated motor systems to handle high-torque faceplate opening and closing routines seamlessly.
- Implemented multi-modal operational support utilizing synchronized Voice Command controls, manual toggles, and mobile app connectivity.

ResQLink

Disaster Communication Platform

Technologies: React, WebRTC, Peer-to-Peer Networking, JavaScript

- Architected a resilient peer-to-peer communication solution designed to operate autonomously during full telecommunication infrastructure failures.
- Utilized decentralized mesh mechanics to enable critical data broadcast channels, localized emergency alerts, and community response logging.
- Built an intuitive, live tactical disaster tracking dashboard user interface optimized for field deployment stability.

Gestyxra

Gesture Controlled 3D Modeling Platform

Technologies: Computer Vision, OpenCV, AI Models, Three.js

- Developed a touchless spatial environment enabling fluid digital 3D model configuration via real-time geometric hand tracking.
- Leveraged computer vision pipelines to isolate and interpret spatial hand coordinate changes without physical hardware interaction.
- Configured responsive three-dimensional matrix manipulation controls including smooth model zooming, translation, and rotational updates.

Technologies: AI, Machine Learning, Computer Vision, Face Recognition Models

- Engineered an automated attendance tracking platform utilizing facial feature clustering structures to scan large classrooms concurrently.
- Designed a centralized management console interface to instantly record, graph, and process individual classroom metric logs.
- Optimized frame recognition latency to achieve highly accurate identification records under varying ambient illumination settings.